

Lecture 5: Stochastic processes & Comparison inequalities

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1 Stochastic processes

Definition 1 (Stochastic process). A stochastic (or random) process is a collection of random variables $(X_t)_{t \in T}$ indexed by time t in a given set T .

A typical example of a stochastic process that is extremely important in the field of data science is the Gaussian process. We give a formal definition below.

Definition 2 (Gaussian process). A stochastic process $(X_t)_{t \in T}$ is a Gaussian process if for any subset $T_0 \subset T$, the random vector $(X_t)_{t \in T_0}$ has a normal distribution.

A classic example of a Gaussian process is Brownian motion, which we will briefly discuss next. Other well-known examples of stochastic processes that we won't be discussing are Bernoulli or Poisson processes.

1.1 Brownian motion

Brownian motion denoted by $(W_t)_{t \in T}$ is a continuous-time stochastic process, characterized by four properties:

- i) $W_0 = 0$
- ii) W_t is almost surely continuous, i.e.

$$\lim_{\delta t \rightarrow 0} \frac{P(|W(t + \delta t) - W(t)| \geq \epsilon)}{\delta t} = 0 \quad \forall \epsilon > 0 \quad (1)$$

- iii) W_t has independent (or stationary) increments

- iv) $W_t - W_s \sim \mathcal{N}(0, t - s)$ for $0 \leq s \leq t$.

Note that second property is expressed with probabilities: dis-continuities in sample functions can only occur with probability zero. Also note that another way to express iii) is $W_t - W_s | \mathcal{F}_s \sim \mathcal{N}(0, t - s)$ (where $\mathcal{F}_s$ is the σ -algebra of W_s).

Figure 1 shows different sample paths of Brownian motion in the 1 dimensional case. If one were to average many sample paths, we would see that the average path is constant at zero.

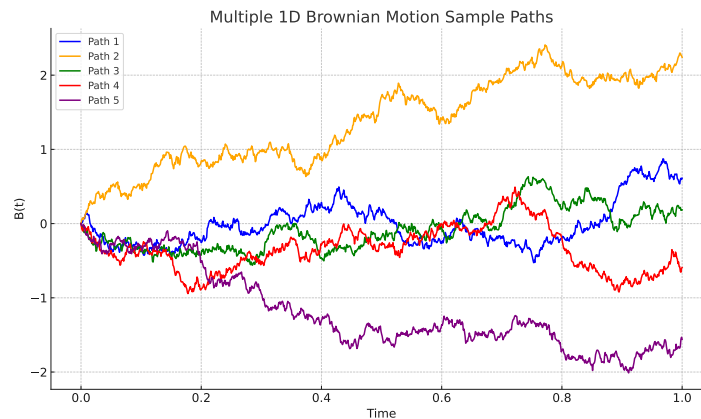


Figure 1: Visualization of Brownian paths. Each path with its color is one path of Brownian motion.

Next, we present a key theorem on Brownian motion increments: for any time interval Δt , both their expectation and their variance grow linearly with Δt .

Theorem 1 (Normally distributed increments of Brownian motion with a drift). *Let $(X(t))_{t \geq 0}$ be a Brownian motion with drift μ and volatility σ , meaning*

$$X(t) = \mu t + \sigma W(t),$$

where $(W(t))_{t \geq 0}$ is a standard Brownian motion. Then

$$X(t) - X(0) \sim \mathcal{N}(\mu t, \sigma^2 t).$$

Proof. We have

$$X(t) - X(0) = (\mu t + \sigma W(t)) - (\mu \cdot 0 + \sigma W(0)) = \mu t + \sigma W(t).$$

Since $W(t) \sim \mathcal{N}(0, t)$, multiplying by σ scales the variance:

$$\sigma W(t) \sim \mathcal{N}(0, \sigma^2 t).$$

Adding a constant μt shifts the mean:

$$\mu t + \sigma W(t) \sim \mathcal{N}(\mu t, \sigma^2 t).$$

Thus,

$$X(t) - X(0) \sim \mathcal{N}(\mu t, \sigma^2 t).$$

□

As a result of this theorem, we have the following density function of a Brownian motion with drift μt :

$$f_{X(t)}(x) = \frac{1}{\sqrt{2\pi\sigma^2 t}} e^{-\frac{(x-\mu t)^2}{2\sigma^2 t}} \quad (2)$$

Note that a property of Brownian motion is that its sample paths are not differentiable which can be easily verified in the mean-square sense:

$$\mathbb{E} \left[\left(\frac{W(t+\delta t) - W(t)}{\delta t} \right)^2 \right] = \frac{\mathbb{E}[(W(t+\delta t) - W(t))^2]}{\delta t^2} = \frac{\sigma^2}{\delta t} \quad (3)$$

This diverges for $\delta t \rightarrow 0$ and therefore it is **not differentiable in L_2** . This is also the case for almost sure convergence.

Example (discrete case). Consider a sequence of steps. At each step, we flip a coin and we go left if the coin is head and right otherwise. We denote by z_k the value of the coin at step k , i.e.

$$z_k = \begin{cases} 1, & p = \frac{1}{2} \\ -1, & p = \frac{1}{2}. \end{cases} \quad (4)$$

Let $x_t = \sum_{k=1}^t z_k$, where t is the time elapsed since the start of the random walk, then the following holds:

$$\mathbb{E}(z_k) = 0 \quad \mathbb{E}(x_t) = 0 \quad (5)$$

$$\mathbb{E}(z_k^2) = \sigma^2 \quad \mathbb{E}(x_t) = \sum_{k=1}^t \mathbb{E}(z_k^2) = t\sigma^2, \quad (6)$$

therefore $\mathbb{E}(x_t)$ is independent of the number of discrete steps, but it increases with time t .

Markov property For $c > 0$, $X(t) := W(t+c) - W(c)$ is a standard BM and it is independent from $\{W(s) : 0 \leq s \leq c\}$.

The strong Markov property is much stronger as it changes the constant c by a random stopping time. For a stopping time τ , $X(\tau) := W(t+\tau) - W(\tau)$ is a standard BM and it is independent of all possible paths of $\{W(s) : 0 \leq s \leq \tau\}$.

Reflection principle The reflection principle is based on an equation (which we will call reflection equality) that states that if the path of a Brownian motion $W(t)$ reaches a value a at time $s \leq t$, then the subsequent path after time s has the same distribution as the reflection of the subsequent path about the value a . We will illustrate this principle in Figure 2.

Theorem 2 (Reflection principle). *If $\{W(t) : t \geq 0\}$ is a Wiener process, and $a > 0$ is a threshold (also called a crossing point), then*

$$P \left(\sup_{0 \leq s \leq t} W(s) \geq a \right) = 2P(W(t) \geq a) \quad (7)$$

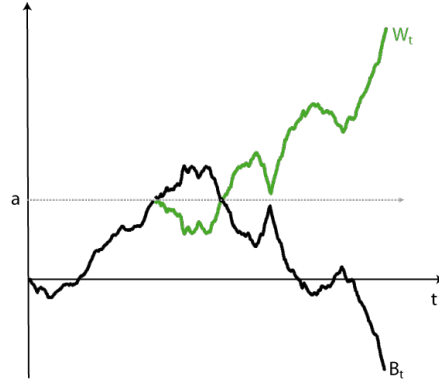


Figure 2: Reflection equality. Once the Brownian motion reaches the value a , then the probability distribution of the black path and the reflected green path are equal.

Source: <https://math.stackexchange.com/questions/3293237/interpretation-reflection-principle>

Proof. Define $M(t) = \max_{0 \leq s \leq t} W(s)$, i.e. the maximum of BM up to time t . We want to compute $P(M(t) \geq a)$ for some $a > 0$.

Note that

$$P(W(t) \geq a) = P(W(t) \geq a, M(t) \geq a) + P(W(t) \geq a, M(t) < a). \quad (8)$$

The second event is impossible, thus it has probability zero. Next, we have

$$P(W(t) \geq a, M(t) \geq a) = P(W(t) \geq a \mid M(t) \geq a)P(M(t) \geq a). \quad (9)$$

The event $\{M(t) \geq a\}$ means that $W(t)$ has reached a at some random time before t .

Now comes a crucial observation: once the Wiener process reached threshold a , then due to the symmetry there is an equal probability ($p = 0.5$) it will end up above or below threshold a at any future time¹. Indeed,

$$P(W(t) \geq a \mid M(t) \geq a) = P(W(t) < a \mid M(t) \geq a) \quad (10)$$

and

$$P(W(t) \geq a \mid M(t) \geq a) + P(W(t) < a \mid M(t) \geq a) = 1, \quad (11)$$

implies that $P(W(t) \geq a \mid M(t) \geq a) = \frac{1}{2}$.

We now obtain

$$P(W(t) \geq a) = \frac{1}{2}P(M(t) \geq a). \quad (12)$$

□

2 Slepian's inequality

In this section, we will study the expected supremum of a stochastic process $\mathbb{E} \sup_{t \in T} X_t$. This object plays an important role in many problems in statistics and learning theory, among others. In the case of Brownian motion, one can show that (see exercise sheet):

$$\mathbb{E} \sup_{t \leq t_0} X_t = \sqrt{\frac{2t_0}{\pi}}, \quad t_0 \geq 0.$$

Calculating a closed-form expression for $\mathbb{E} \sup_{t \in T} X_t$ for arbitrary processes X_t is not always possible. However, a general bound is provided by Slepian's inequality. To present this inequality, we will more specifically consider the case where T is finite. Then the process $(X_t)_{t \in T}$ is a Gaussian vector in $\mathbb{R}^d$ where $d = |T|$. This implies that we will be interested in the quantity $\max_{i \in [d]} X_i$.

To build some intuition, let's first work through a simple example before stating the theorem. Consider two d -dimensional random processes $X \sim \mathcal{N}(0, \mathbf{I}_d)$ and $Y \sim \mathcal{N}(0, \Sigma_d)$ where the covariance Σ_d is such that

$$\mathbb{E} Y_i Y_j = \begin{cases} 1 & i = j \\ \frac{1}{2d} & i \neq j. \end{cases}$$

Which process do you expect will have a larger maximum? Note that the covariance matrix of Y is such that it exhibits positive correlation, i.e. variables are more closely related. Therefore, we expect the maximum to be smaller.

¹This argument relies on the strong Markov property: after a random time T_a , $W(s + T_a) - W(T_a)$ is independent of whatever happens before T_a .

Let's now introduce the main result of this section.

Theorem 3 (Slepian's inequality). *Consider two mean-zero GPs $(X_t)_{t \in T}$ and $(Y_t)_{t \in T}$ such that*

$$\mathbb{E}X_t^2 = \mathbb{E}Y_t^2 \text{ and } \mathbb{E}(X_t - X_s)^2 \leq \mathbb{E}(Y_t - Y_s)^2.$$

Then, $\forall \tau \in \mathbb{R}$, $\Pr(\max_{i \in [d]} X_i \geq \tau) \leq \Pr(\max_{i \in [d]} Y_i \geq \tau)$. Consequently, $\mathbb{E} \max_{i \in [d]} X_i \leq \mathbb{E} \max_{i \in [d]} Y_i$.

Informally, Slepian's inequality states that among two mean-zero Gaussian processes with the same point-wise variance, the one whose covariances between distinct points are smaller will have the larger expected supremum. The remark below demonstrates how one can interpret the condition of the theorem as a comparison of the covariances of the two processes.

Remark 1. *Since both processes are centered and*

$$\text{var}(X_t) = \mathbb{E}[X_t^2] = \mathbb{E}[Y_t^2] = \text{var}(Y_t) = \sigma^2,$$

we can write

$$\mathbb{E}(X_t - X_s)^2 = \text{var}(X_t) + \text{var}(X_s) - 2 \text{Cov}(X_t, X_s) = 2\sigma^2 - 2 \text{Cov}(X_t, X_s),$$

and similarly

$$\mathbb{E}(Y_t - Y_s)^2 = 2\sigma^2 - 2 \text{Cov}(Y_t, Y_s).$$

The condition

$$\mathbb{E}(X_t - X_s)^2 \leq \mathbb{E}(Y_t - Y_s)^2$$

then implies

$$2\sigma^2 - 2 \text{Cov}(X_t, X_s) \leq 2\sigma^2 - 2 \text{Cov}(Y_t, Y_s) \implies \text{Cov}(X_t, X_s) \geq \text{Cov}(Y_t, Y_s) \quad \forall s, t \in T.$$

A related result to Slepian's inequality is the Sudakov–Fernique inequality, which we state here without proof.

Theorem 4 (Sudakov–Fernique inequality). *Let $(X_t)_{t \in T}$ and $(Y_t)_{t \in T}$ be two mean-zero Gaussian processes. Assume that, for all $t, s \in T$, we have*

$$\mathbb{E}(X_t - X_s)^2 \leq \mathbb{E}(Y_t - Y_s)^2.$$

Then

$$\mathbb{E}[\sup_{t \in T} X_t] \leq \mathbb{E}[\sup_{t \in T} Y_t].$$

Proof of Slepian's inequality

We start with some helper lemmas.

Lemma 5 (Stein's lemma). *Let $X \sim \mathcal{N}(\mu, \sigma^2)$ and suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is a function for which the two expectations $\mathbb{E}[f(X)(X - \mu)]$ and $\mathbb{E}[f'(X)]$ both exist. Then*

$$\mathbb{E}[f(X)(X - \mu)] = \sigma^2 \mathbb{E}[f'(X)].$$

Proof. We provide a proof for the case where $\sigma^2 = 1$. The case of a general variance σ^2 follows by substitution.

Recall that the probability density function for the normal distribution with expectation 0 and variance 1 is

$$p(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}.$$

Since $\int x \exp(-x^2/2) = -\exp(-x^2/2)$ we get from integration by parts ($\int u dv = uv - \int v du$):

$$\mathbb{E}[g(X)X] = \frac{1}{\sqrt{2\pi}} \int g(x)x \exp(-x^2/2) dx = -\frac{1}{\sqrt{2\pi}} \int g'(x) \exp(-x^2/2) dx = \mathbb{E}[g'(X)].$$

Note that, when using integrations by parts, the uv term cancels since:

- $e^{-x^2/2} \rightarrow 0$ faster than any polynomial as $|x| \rightarrow \infty$.

- $g(x)$ is such that $\mathbb{E}[g(X)X]$ exists, meaning $g(x)e^{-x^2/2}$ must be integrable.

These two facts imply

$$\lim_{x \rightarrow \pm\infty} g(x) e^{-x^2/2} = 0.$$

Hence

$$[-g(x)e^{-x^2/2}]_{-\infty}^{\infty} = 0.$$

□

Lemma 6 ((Kahane, 1986)). *Let $X \sim \mathcal{N}(m_x, \Sigma^x)$ and $Y \sim \mathcal{N}(m_y, \Sigma^y)$ be d multivariate Gaussian vectors such that $\mathbb{E}[X_i] = \mathbb{E}[Y_i]$ for all $i \in [d]$. Let $A = \{(i, j) : \Sigma_{ij}^x < \Sigma_{ij}^y\}$ and $B = \{(i, j) : \Sigma_{ij}^x > \Sigma_{ij}^y\}$. Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be any twice differentiable function with partial derivatives having subgaussian growth and such that $\partial_i \partial_j f \geq 0$ for all $(i, j) \in A$ and $\partial_i \partial_j f \leq 0$ for all $(i, j) \in B$. Then, $\mathbb{E}[f(X)] \leq \mathbb{E}[f(Y)]$.*

Proof. We define an interpolator between X and Y as

$$Z(t) := \sqrt{t}X + \sqrt{1-t}Y, \quad \text{where } t \in [0, 1].$$

Note that $Z(0) = Y$ and $Z(1) = X$ and

$$\mathbb{E}[f(Y) - f(X)] = \mathbb{E} \left[\int_0^1 \frac{d}{dt} f(Z(t)) dt \right] = \int_0^1 \frac{d}{dt} \mathbb{E}[f(Z(t))] dt,$$

where, under some mild conditions on f (continuous and bounded), we exchanged the expectation and the integral.

Taking the derivative (using the multivariate chain rule), we obtain

$$\begin{aligned} \frac{d}{dt} \mathbb{E}f(Z(t)) &= \sum_{i=1}^d \mathbb{E} \frac{\partial f}{\partial Z_i}(Z(t)) \frac{dZ_i}{dt} \\ &= \frac{1}{2} \sum_{i=1}^d \mathbb{E} \frac{\partial f}{\partial Z_i}(Z(t)) \left(\frac{X_i}{\sqrt{t}} - \frac{Y_i}{\sqrt{1-t}} \right) \end{aligned} \quad (13)$$

We first focus on the first term in the wide brackets. Let $g_i(Z) := \frac{\partial f}{\partial Z_i}(Z(t))$ such that we can use the more compact notation $\sum_{i=1}^d \frac{1}{\sqrt{t}} \mathbb{E} X_i \frac{\partial f}{\partial Z_i}(Z(t)) = \sum_{i=1}^d \frac{1}{\sqrt{t}} \mathbb{E} X_i g_i(Z)$.

By conditioning on Y and applying Stein's lemma, we get

$$\begin{aligned} \mathbb{E}[X_i g_i(X) | Y] &= \sum_{j=1}^d \Sigma_{ij}^y \mathbb{E} \frac{\partial g_i}{\partial Z_j}(X) \\ &= \sum_{j=1}^d \Sigma_{ij}^y \mathbb{E} \frac{\partial^2 f}{\partial Z_i \partial Z_j}(Z(t)) \sqrt{t}. \end{aligned}$$

Taking expectation with respect to Y , we obtain

$$\sum_{i=1}^d \frac{1}{\sqrt{t}} \mathbb{E} X_i \frac{\partial f}{\partial Z_i}(Z(t)) = \sum_{i,j=1}^d \Sigma_{ij}^y \mathbb{E} \frac{\partial^2 f}{\partial Z_i \partial Z_j}(Z(t)).$$

Similar steps can be followed for the second term in the wide bracket of Eq. (13), which yields

$$\frac{d}{dt} \mathbb{E}f(Z(t)) = \frac{1}{2} \sum_{i,j=1}^d (\Sigma_{ij}^x - \Sigma_{ij}^y) \mathbb{E} \frac{\partial^2 f}{\partial Z_i \partial Z_j}(Z(t)).$$

Using the assumptions, we conclude that

$$\frac{d}{dt} \mathbb{E}f(Z(t)) \geq 0,$$

which implies that $\mathbb{E}f(Z(t))$ increases with t , i.e. $\mathbb{E}f(X) \geq \mathbb{E}f(Y)$.

□

We are now ready to prove Slepian's inequality.

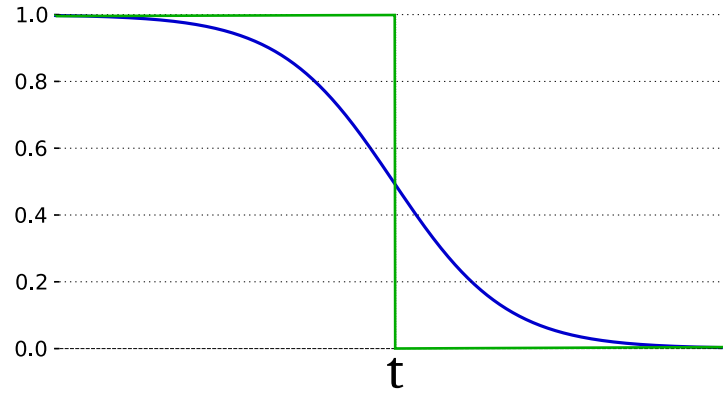


Figure 3: Smooth approximation $\phi_\epsilon(x)$ (shown in blue) to step function (shown in green).

Proof Slepian's inequality. Note that $\Pr(\max_{i \in [d]} X_i \geq \tau) = \mathbb{E}[\mathbf{1}_{\max_{i \in [d]} X_i \geq \tau}]$ so we would be tempted to apply Kahane's lemma with $f(\max_{i \in [d]} X_i) = \mathbb{E}[\mathbf{1}_{\max_{i \in [d]} X_i \geq \tau}]$. However, it requires a C^2 function. The general idea will therefore be to construct a C^2 function that approximates the indicator function.

To do so, we define a function $\phi_\epsilon(x)$ shown in Figure 3 below.

This function is constructed so that

$$\phi_\epsilon(x) = \begin{cases} 1 & x < t - \epsilon \\ 0 & x \geq t + \epsilon, \end{cases}$$

and smoothly interpolates in the interval $[t - \epsilon, t + \epsilon]$.

It is therefore smooth and non-increasing (i.e. $\phi'_\epsilon(x) \leq 0$) on the entire domain. We then define $g_\epsilon(x) := \prod_{i=1}^d \phi_\epsilon(x_i)$ to smoothly approximate the function $\max_{i \in [d]} X_i < t$. Note that as $\epsilon \rightarrow 0$, $g_\epsilon(x) \rightarrow \mathbf{1}_{\max_{i \in [d]} X_i < t}$ (note we are approximating $\mathbf{1}_{\max_{i \in [d]} X_i < t}$, not $\mathbf{1}_{\max_{i \in [d]} X_i \geq t}$).

Based on the covariance assumption stated in Slepian's inequality ($\Sigma^x < \Sigma^y$), we require that $\partial_i \partial_j g \geq 0$ in order to apply Kahane's lemma. This condition is clearly satisfied since:

$$\partial_i \partial_j g_\epsilon(x) = \underbrace{\phi'_\epsilon(x_i)}_{\leq 0} \underbrace{\phi'_\epsilon(x_j)}_{\leq 0} \prod_{k \neq i, j} \underbrace{\phi_\epsilon(x_k)}_{\geq 0} \geq 0.$$

Applying Kahane's lemma yields

$$\mathbb{E}[g_\epsilon(x)] \geq \mathbb{E}[g_\epsilon(y)].$$

By the dominated convergence theorem, this implies that

$$\mathbb{E}[\mathbf{1}_{\max_{i \in [d]} X_i < t}] \geq \mathbb{E}[\mathbf{1}_{\max_{i \in [d]} Y_i < t}],$$

i.e.

$$\mathbb{E}[\mathbf{1}_{\max_{i \in [d]} X_i \geq t}] \leq \mathbb{E}[\mathbf{1}_{\max_{i \in [d]} Y_i \geq t}].$$

□

3 Bounds for Gaussian Random Matrices

Our next goal is to demonstrate how the Gaussian comparison principles yield tight estimates on the spectral norm of a random matrix. Recall from an earlier lecture that if $\mathbf{A}$ is an $m \times n$ matrix whose rows are independent sub-Gaussian vectors, then an ϵ -net argument shows

$$\mathbb{E} \|\mathbf{A}\| \leq \sqrt{m} + C \sqrt{n},$$

for some absolute constant C . However, if the entries of $\mathbf{A}$ are Gaussian, the Sudakov–Fernique inequality refines this bound to its optimal version with $C = 1$.

Theorem 7 (Norms of Gaussian Random Matrices). *Let $\mathbf{A}$ be an $m \times n$ matrix with i.i.d. $\mathcal{N}(0, 1)$ entries. Then*

$$\mathbb{E} \|\mathbf{A}\| \leq \sqrt{m} + \sqrt{n}.$$

Proof. Observe that

$$\|\mathbf{A}\| = \sup_{\mathbf{u} \in S^{n-1}, \mathbf{v} \in S^{m-1}} \langle \mathbf{A}\mathbf{u}, \mathbf{v} \rangle = \sup_{(\mathbf{u}, \mathbf{v}) \in S^{n-1} \times S^{m-1}} X_{uv},$$

where T denotes the product set $\mathcal{S}^{n-1} \times \mathcal{S}^{m-1}$ and

$$X_{uv} := \langle \mathbf{A}\mathbf{u}, \mathbf{v} \rangle \sim \mathcal{N}(0, 1).$$

(Check!) To apply the Sudakov–Fernique comparison inequality (Theorem 4), let us compute the increments of the process (X_{uv}) . For any $(\mathbf{u}, \mathbf{v}), (\mathbf{w}, \mathbf{z}) \in T$, we have

$$\begin{aligned} \mathbb{E}(X_{uv} - X_{wz})^2 &= \mathbb{E}(\langle \mathbf{A}\mathbf{u}, \mathbf{v} \rangle - \langle \mathbf{A}\mathbf{w}, \mathbf{z} \rangle)^2 = \mathbb{E}\left(\sum_{i,j} A_{ij}(u_j v_i - w_j z_i)\right)^2 \\ &= \sum_{i,j} (u_j v_i - w_j z_i)^2 \quad (\text{by independence, mean 0, variance 1}) \\ &= \|\mathbf{u}\mathbf{v}^\top - \mathbf{w}\mathbf{z}^\top\|_F^2 \\ &\leq \|\mathbf{u} - \mathbf{w}\|_2^2 + \|\mathbf{v} - \mathbf{z}\|_2^2 \quad (\text{see Exercise}). \end{aligned}$$

Next, define a simpler Gaussian process (Y_{uv}) with the same increment structure:

$$Y_{uv} := \langle \mathbf{g}, \mathbf{u} \rangle + \langle \mathbf{h}, \mathbf{v} \rangle, \quad (\mathbf{u}, \mathbf{v}) \in T,$$

where

$$\mathbf{g} \sim \mathcal{N}(0, \mathbf{I}_n), \quad \mathbf{h} \sim \mathcal{N}(0, \mathbf{I}_m)$$

are independent. Its increments satisfy

$$\begin{aligned} \mathbb{E}(Y_{uv} - Y_{wz})^2 &= \mathbb{E}(\langle \mathbf{g}, \mathbf{u} - \mathbf{w} \rangle + \langle \mathbf{h}, \mathbf{v} - \mathbf{z} \rangle)^2 \\ &= \mathbb{E}\langle \mathbf{g}, \mathbf{u} - \mathbf{w} \rangle^2 + \mathbb{E}\langle \mathbf{h}, \mathbf{v} - \mathbf{z} \rangle^2 \quad (\text{by independence, mean 0}) \\ &= \|\mathbf{u} - \mathbf{w}\|_2^2 + \|\mathbf{v} - \mathbf{z}\|_2^2 \quad (\text{since } \mathbf{g}, \mathbf{h} \text{ are standard normal}). \end{aligned}$$

To prove the last equality, consider $\mathbf{g} \sim \mathcal{N}(0, \mathbf{I}_n)$ and $\mathbf{x} \in \mathbb{R}^n$, then $\langle \mathbf{g}, \mathbf{x} \rangle = \sum_{i=1}^n g_i x_i$, and since each $g_i \sim \mathcal{N}(0, 1)$ and the coordinates are independent,

$$\mathbb{E}(g_i x_i)^2 = x_i^2, \quad \mathbb{E}(g_i x_i g_j x_j) = 0 \quad \text{for } i \neq j.$$

Thus

$$\mathbb{E}\langle \mathbf{g}, \mathbf{x} \rangle^2 = \sum_{i=1}^n x_i^2 = \|\mathbf{x}\|_2^2.$$

Comparing the two processes, we see that

$$\mathbb{E}(X_{uv} - X_{wz})^2 \leq \mathbb{E}(Y_{uv} - Y_{wz})^2 \quad \text{for all } (\mathbf{u}, \mathbf{v}), (\mathbf{w}, \mathbf{z}) \in T,$$

as required by Sudakov–Fernique. Applying Theorem 4 gives

$$\begin{aligned} \mathbb{E}\|\mathbf{A}\| &= \mathbb{E} \sup_{(\mathbf{u}, \mathbf{v}) \in T} X_{uv} \leq \mathbb{E} \sup_{(\mathbf{u}, \mathbf{v}) \in T} Y_{uv} \\ &= \mathbb{E} \sup_{\mathbf{u} \in \mathcal{S}^{n-1}} \langle \mathbf{g}, \mathbf{u} \rangle + \mathbb{E} \sup_{\mathbf{v} \in \mathcal{S}^{m-1}} \langle \mathbf{h}, \mathbf{v} \rangle \\ &= \mathbb{E}\|\mathbf{g}\|_2 + \mathbb{E}\|\mathbf{h}\|_2 \\ &\leq (\mathbb{E}\|\mathbf{g}\|_2^2)^{1/2} + (\mathbb{E}\|\mathbf{h}\|_2^2)^{1/2} \quad (\text{by Cauchy-Schwarz}) \\ &= \sqrt{n} + \sqrt{m}. \end{aligned}$$

□

References

Jean-Pierre Kahane. Une inégalité du type de slepian et gordon sur les processus gaussiens. *Israel Journal of Mathematics*, 55(1): 109–110, 1986.